

Estimation of error components models with the plm function

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`plm` is a very versatile function which enable the estimation of a wide range of error component models. Those models can be written as follows :

$$y_{nt} = \alpha + \beta^T x_{nt} + \epsilon_{nt} = \alpha + \beta^T x_{nt} + \eta_n + \mu_t + v_{nt}$$

where n and t are the individual and time indexes, y the response, x a vector of covariates, α the overall intercept and β the vector of parameters of interest that we are willing to estimate. The error term ϵ_{nt} is composed of three elements (in the two-way case):

- η_n is the individual effect,
- μ_t is the time effect,
- v_{nt} is the idiosyncratic error.

Basic use of `plm`

The first two arguments of `plm` are, like for most of the estimation functions of R a `formula` which describes the model to be estimated and a `data.frame`. `subset`, `weights`, and `na.action` are also available and have the same behavior as in the `lm` function. Three more main arguments can be set :

- `index` helps `plm` to understand the structure of the data : if `NULL`, the first two columns of the data are assumed to contain the individual or the time index. Otherwise, supply the column names of the individual and time index as a character, e.g., use something like `c("firm", "year")` or just `"firm"` if there is no explicit time index.
- `effect` indicates the effects that should be taken into account ; this is one of `"individual"`, `"time"`, and `"twoways"`.
- `model` indicates the model to be estimated :
 - `"pooling"` is the OLS estimation (equivalent to a call to `lm`),
 - `"between"` performs the estimation on the individual or time means,
 - `"within"` (fixed-effects model) on the deviations from the individual or/and time mean,
 - `"fd"` on the first differences, and
 - `"random"` (random-effects model) perform a feasible generalized least squares estimation which takes into account the correlation induced by the presence of individual and/or time effects.

The estimation of all but the last model is straightforward, as it requires only the estimation by *OLS* of obvious transformations of the data. The *GLS* model requires more explanation. In most of the cases, the estimation is obtained by quasi-differencing the data from the individual and/or the time means. The coefficients used to perform this quasi-difference depends on estimators of the variance of the components of the error, namely σ_v^2 , σ_η^2 in case of individual effects and σ_μ^2 in case of time effects.

The most common technique used to estimate these variance is to use the following result :

$$\frac{E(\epsilon^T W \epsilon)}{N(T-1)} = \sigma_v^2$$

and

$$\frac{E(\epsilon^T B \epsilon)}{N} = T\sigma_\eta^2 + \sigma_v^2$$

where B and W are respectively the matrices that performs the individual (or time) means and the deviations from these means. Consistent estimators can be obtained by replacing the unknown errors by the residuals of a consistent preliminary estimation and by dropping the expecting value operator. Some degree of freedom correction can also be introduced. `plm` calls the general function `ercomp` to estimate the variances. Important arguments to `ercomp` are:

- `models` indicates which models are estimated in order to calculate the two quadratic forms ; for example `c("within", "Between")`. Note that when only one model is provided in `models`, this means that the same residuals are used to compute the two quadratic forms.
- `dfcor` indicates what kind of degrees of freedom correction is used : if 0, the quadratic forms are divided by the number of observations, respectively $N \times T$ and N ; if 1, the numerators of the previous expressions are used ($N \times (T - 1)$ and N) ; if 2, the number of estimated parameters in the preliminary estimate K is deducted. Finally, if 3, the unbiased version is computed, which is based on much more complex computations, which relies on the calculus of the trace of different cross-products which depends on the preliminary models used.
- `method` is an alternative to the `models` argument; it is one of :
 - "walhus" (equivalent to setting `models = c("pooling")`), Wallace and Hussain (1969),
 - "swar" (equivalent to `models = c("within", "Between")`), Swamy and Arora (1972),
 - "amemiya" (equivalent to `models = c("within")`), T. Amemiya (1971),
 - "nerlove", which is a specific method which doesn't fit to the quadratic form methodology described above (Nerlove (1971)) and uses an within model for the variance estimation as well,
 - "ht" is an slightly modified version of "amemiya": when there are time-invariant covariates, the T. Amemiya (1971) estimator of the individual component of the variance underestimates as the time-invariant covariates disappear in the within regression. In this case, Hausman and Taylor (1981) proposed to regress the estimation of the individual effects on the time-invariant covariates and use the residuals in order to estimate the components of the variance.

Note that for `plm`, the arguments are `random.models`, `random.dfcor`, and `random.method` and correspond to arguments `models`, `method`, and `random.dfcor` of function `ercomp` with the same values as above, respectively.

To illustrate the use of `plm`, we use examples reproduced in B. H. Baltagi (2013), p. 21; B. H. Baltagi (2021), p. 31, table 2.1 presents EViews' results of the estimation on the Grunfeld data set :

```
library("plm")
data("Grunfeld", package = "plm")
ols <- plm(inv ~ value + capital, Grunfeld, model = "pooling")
between <- update(ols, model = "between")
within <- update(ols, model = "within")
walhus <- update(ols, model = "random", random.method = "walhus", random.dfcor = 3)
amemiya <- update(walhus, random.method = "amemiya")
swar <- update(amemiya, random.method = "swar")
```

Note that the `random.dfcor` argument is set to 3, which means that the unbiased version of the estimation of the error components is used. We use the `texreg` package to present the results :

```
library("texreg")
screenreg(list(ols = ols, between = between, within = within,
              walhus = walhus, amemiya = amemiya, swar = swar),
          digits = 5, omit.coef = "(Intercept)")
```

```
##
##
=====
##          ols          between    within      walhus      amemiya      swar
## -----
## value      0.11556 ***    0.13465 **    0.11012 ***    0.10979 ***    0.10978 ***
0.10978 ***
##          (0.00584)      (0.02875)    (0.01186)    (0.01052)    (0.01048)
(0.01049)
## capital    0.23068 ***    0.03203      0.31007 ***    0.30818 ***    0.30808 ***
0.30811 ***
##          (0.02548)      (0.19094)    (0.01735)    (0.01717)    (0.01718)
(0.01718)
## -----
## R^2        0.81241      0.85777      0.76676      0.76941      0.76954
0.76950
## Adj. R^2    0.81050      0.81713      0.75311      0.76707      0.76720
0.76716
## Num. obs.  200          10          200          200          200          200
## s_idios      52.76797
52.76797
## s_id        84.20095
84.20095
##
##
=====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

The estimated variance can be extracted using the `ercomp` function. For example, for the `amemiya` model :

```
ercomp(amemiya)
```

```
##          var std.dev share
## idiosyncratic 2784.46  52.77 0.285
## individual    6976.18  83.52 0.715
## theta: 0.8601
```

B. H. Baltagi (2013), p. 27; B. H. Baltagi (2021), p. 31 presents the Stata estimation of the Swamy-Arora estimator ; the Swamy-Arora estimator is the same if `random.dfcor` is set to 3 or 2 (the quadratic forms are divided by $\sum_n T_n - K - N$ and by $N - K - 1$), so I don't know what is the behaviour of Stata for the other estimators for which the unbiased estimators differs from the simple one.

```
data("Produc", package = "plm")
PrSwar <- plm(log(gsp) ~ log(pcap) + log(pc) + log(emp) + unemp, Produc,
             model = "random", random.method = "swar", random.dfcor = 3)
summary(PrSwar)
```

```
## Oneway (individual) effect Random Effect Model
## (Swamy-Arora's transformation)
##
## Call:
```

```
## plm(formula = log(gsp) ~ log(pcap) + log(pc) + log(emp) + unemp,
##       data = Produc, model = "random", random.method = "swar",
##       random.dfcor = 3)
##
## Balanced Panel: n = 48, T = 17, N = 816
##
## Effects:
##               var   std.dev share
## idiosyncratic 0.001454 0.038137 0.175
## individual    0.006838 0.082691 0.825
## theta: 0.8888
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.1067230 -0.0245520 -0.0023694  0.0217333  0.1996307
##
## Coefficients:
##              Estimate Std. Error z-value      Pr(>|z|)
## (Intercept)  2.13541100  0.13346149 16.0002 < 0.00000000000000022 ***
## log(pcap)    0.00443859  0.02341732  0.1895      0.8497
## log(pc)      0.31054843  0.01980475 15.6805 < 0.00000000000000022 ***
## log(emp)     0.72967053  0.02492022 29.2803 < 0.00000000000000022 ***
## unemp        -0.00617247  0.00090728 -6.8033  0.0000000001023 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    29.209
## Residual Sum of Squares: 1.1879
## R-Squared:    0.95933
## Adj. R-Squared: 0.95913
## Chisq: 19131.1 on 4 DF, p-value: < 0.00000000000000022
```

The twoways effect model

The two-ways effect model is obtained by setting the `effect` argument to "twoways". B. H. Baltagi (2013) pp. 51-53; B. H. Baltagi (2021), pp. 61-62, tables 3.1-3.3, presents EViews' output for the Grunfeld data set.

```
Grw <- plm(inv ~ value + capital, Grunfeld, model = "random", effect = "twoways",
           random.method = "walhus", random.dfcor = 3)
Grs <- update(Grw, random.method = "swar")
Gra <- update(Grw, random.method = "amemiya")
screenreg(list("Wallace-Hussain" = Grw, "Swamy-Arora" = Grs, "Amemiya" = Gra), digits = 5)

##
## =====
##              Wallace-Hussain  Swamy-Arora  Amemiya
## -----
## (Intercept)  -57.81705 *      -57.86538 *      -63.89217 *
##              (28.63258)      (29.39336)      (30.53284)
## value        0.10978 ***      0.10979 ***      0.11145 ***
##              (0.01047)      (0.01053)      (0.01096)
## capital      0.30807 ***      0.30819 ***      0.32353 ***
```

```
##          (0.01719)      (0.01717)      (0.01877)
## -----
## s_idios      55.33298      51.72452      51.72452
## s_id         87.31428      84.23332      89.26257
## s_time        0.00000      0.00000      15.77783
## R^2           0.76956      0.76940      0.74898
## Adj. R^2       0.76722      0.76706      0.74643
## Num. obs.     200          200          200
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

The estimated variance of the time component is negative for the Wallace-Hussain as well as the Swamy-Arora models and `plm` sets it to 0.

B. H. Baltagi (2009) pp. 60-62, presents EViews' output for the `Produc` data.

```
data("Produc", package = "plm")
Prw <- plm(log(gsp) ~ log(pcap) + log(pc) + log(emp) + unemp, Produc,
           model = "random", random.method = "walhus",
           effect = "twoways", random.dfcor = 3)
Prs <- update(Prw, random.method = "swar")
Pra <- update(Prw, random.method = "amemiya")
screenreg(list("Wallace-Hussain" = Prw, "Swamy-Arora" = Prs, "Amemiya" = Pra), digits = 5)
```

```
##
## =====
##          Wallace-Hussain  Swamy-Arora  Amemiya
## -----
## (Intercept)  2.39200 ***    2.36350 ***    2.85210 ***
##              (0.13833)      (0.13891)      (0.18502)
## log(pcap)     0.02562          0.01785          0.00221
##              (0.02336)      (0.02332)      (0.02469)
## log(pc)       0.25781 ***    0.26559 ***    0.21666 ***
##              (0.02128)      (0.02098)      (0.02438)
## log(emp)      0.74180 ***    0.74490 ***    0.77005 ***
##              (0.02371)      (0.02411)      (0.02584)
## unemp        -0.00455 ***   -0.00458 ***   -0.00398 ***
##              (0.00106)      (0.00102)      (0.00108)
## -----
## s_idios       0.03571          0.03429          0.03429
## s_id          0.08244          0.08279          0.15390
## s_time        0.01595          0.00984          0.02608
## R^2           0.92915          0.93212          0.85826
## Adj. R^2      0.92880          0.93178          0.85756
## Num. obs.     816            816            816
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

Unbalanced panels

Two difficulties arise with unbalanced panels :

- There are no obvious denominators for the quadratic forms of the residuals that are used to estimate the components of the variance. The strategy is then to compute the expected value

and equate it to the actual quadratic forms. Detailed formula are omitted here, they depend on the preliminary estimator.

- For the one-way effect model, the estimator is still obtained by applying *OLS* on demeaned data (the individual **and** the time means are now deducted) for the within model and on quasi-demeaned data for the random effects model ; this is not the case for the two-ways effects model.

B. H. Baltagi (2021), B. H. Baltagi (2013), and B. H. Baltagi (2009) present results of the estimation of the Swamy and Arora (1972) model with the `Hedonic` data set. B. H. Baltagi (2013), p. 195; B. H. Baltagi (2021), p. 237, table 9.1, presents the Stata output and B. H. Baltagi (2009), p. 211 presents EViews' output. EViews' Wallace-Hussain estimator is reported in B. H. Baltagi (2009), p. 210.

```
data("Hedonic", package = "plm")
form <- mv ~ crim + zn + indus + chas + nox + rm +
  age + dis + rad + tax + ptratio + blacks + lstat
HedStata <- plm(form, Hedonic, model = "random", index = "townid",
  random.models = c("within", "between"))
HedEviews <- plm(form, Hedonic, model = "random", index = "townid",
  random.models = c("within", "Between"))
HedEviewsWH <- update(HedEviews, random.models = "pooling")
screenreg(list(EViews = HedEviews, Stata = HedStata, "Wallace-Hussain" = HedEviewsWH),
  digits = 5, single.row = TRUE)
```

```
##
## =====
##              EViews              Stata              Wallace-Hussain
## -----
## (Intercept)   9.68587 (0.19751) ***   9.67780 (0.20714) ***   9.68443 (0.19922) ***
## crim         -0.00741 (0.00105) ***   -0.00723 (0.00103) ***   -0.00738 (0.00105) ***
## zn           0.00008 (0.00065)         0.00004 (0.00069)         0.00007 (0.00066)
## indus        0.00156 (0.00403)         0.00208 (0.00434)         0.00165 (0.00409)
## chasyes      -0.00442 (0.02921)         -0.01059 (0.02896)         -0.00565 (0.02916)
## nox          -0.00584 (0.00125) ***   -0.00586 (0.00125) ***   -0.00585 (0.00125) ***
## rm           0.00906 (0.00119) ***   0.00918 (0.00118) ***   0.00908 (0.00119) ***
## age          -0.00086 (0.00047)         -0.00093 (0.00046) *      -0.00087 (0.00047)
## dis          -0.14442 (0.04409) **      -0.13288 (0.04568) **      -0.14236 (0.04439) **
## rad           0.09598 (0.02661) ***      0.09686 (0.02835) ***      0.09614 (0.02692) ***
## tax          -0.00038 (0.00018) *        -0.00037 (0.00019) *        -0.00038 (0.00018) *
## ptratio      -0.02948 (0.00907) **      -0.02972 (0.00975) **      -0.02951 (0.00919) **
## blacks       0.56278 (0.10197) ***      0.57506 (0.10103) ***      0.56520 (0.10179) ***
## lstat        -0.29107 (0.02393) ***      -0.28514 (0.02385) ***      -0.28991 (0.02391) ***
## -----
## s_idios       0.13025                  0.13025                  0.14050
## s_id          0.11505                  0.12974                  0.12698
## R^2           0.99091                  0.99029                  0.99081
## Adj. R^2      0.99067                  0.99004                  0.99057
## Num. obs.     506                      506                      506
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

The difference is due to the fact that Stata uses a between regression on N observations while EViews uses a between regression on $\sum_n T_n$ observations, which are not the same on unbalanced panels. Note the use of between with or without the B capitalized ("Between" and "between") in the `random.models` argument. `plm`'s default is to use the between regression with $\sum_n T_n$ observations when setting `model = "random"`, `random.method = "swar"`. The default employed is what the original paper

for the unbalanced one-way Swamy-Arora estimator defined (in B. H. Baltagi and Chang (1994), p. 73). A more detailed analysis of Stata's Swamy-Arora estimation procedure is given by Cottrell (2017).

Instrumental variable estimators

All of the models presented above may be estimated using instrumental variables (IV). The instruments are specified using two- or three-part formulas, each part being separated by a `|` sign :

- the first part contains the covariates,
- the second part contains the "double-exogenous" instruments, *i.e.*, variables that can be used twice as instruments, using their within and the between transformation,
- the third part contains the "single-exogenous" instruments, *i.e.*, variables for which only the within transformation can be used as instruments, those variables being correlated with the individual effects.

The instrumental variables estimator used is indicated with the `inst.method` argument:

- "bvk", from Balestra and Varadharajan-Krishnakumar (1987), the default value : in this case, all the instruments are introduced in quasi-differences, using the same transformation as for the response and the covariates,
- "baltagi", from B. H. Baltagi (1981), the instruments of the *second* part are introduced twice by using the between and the within transformation and instruments of the *third* part are introduced with only the within transformation,
- "am", from Takeshi Amemiya and MaCurdy (1986), in addition to the instrument set of "baltagi", the within transformation of the variables of the *second* part for each period are also included as instruments,
- "bms", from Breusch, Mizon, and Schmidt (1989), in addition to the instrument set of "baltagi", the within transformation of the variables of the *second* and the *third* part for each period are included as instruments.

The various possible values of the `inst.method` argument are not relevant for fixed effect IV models as there is only one method for this type of IV models but many for random effect IV models.

The instrumental variable estimators are illustrated in the following example from B. H. Baltagi (2005), pp. 117/120; B. H. Baltagi (2013), pp. 133/137; B. H. Baltagi (2021), pp. 162/165, tables 7.1, 7.3.

```
data("Crime", package = "plm")
crbalt <- plm(lcrmrte ~ lprbarr + lpolpc + lprbconv + lprbpris + lavgsen +
             ldensity + lwcon + lwtuc + lwtrd + lwfir + lwser + lwmfg + lwfed +
             lwsta + lwloc + lpctymle + lpctmin + region + smsa + factor(year)
             | . - lprbarr - lpolpc + ltaxpc + lmix,
             data = Crime, model = "random", inst.method = "baltagi")
crbvk <- update(crbalt, inst.method = "bvk")
crwth <- update(crbalt, model = "within")
crbe <- update(crbalt, model = "between")
screenreg(list(FE2SLS = crwth, BE2SLS = crbe, EC2SLS = crbalt, G2SLS = crbvk),
           single.row = FALSE, digits = 5, omit.coef = "(region)|(year)",
           reorder.coef = c(1:16, 19, 18, 17))
```

```
##
## =====
##           FE2SLS      BE2SLS      EC2SLS      G2SLS
## -----
## lprbarr      -0.57551   -0.50294 *    -0.41293 ***   -0.41414
##              (0.80218) (0.24062)      (0.09740)      (0.22105)
## lpolpc        0.65753    0.40844 *    0.43475 ***    0.50495 *
```

```
##          (0.84687) (0.19300) (0.08970) (0.22778)
## lprbconv -0.42314 -0.52477 *** -0.32289 *** -0.34325 **
##          (0.50194) (0.09995) (0.05355) (0.13246)
## lprbpris -0.25026 0.18718 -0.18632 *** -0.19005 **
##          (0.27946) (0.31829) (0.04194) (0.07334)
## lavgsen 0.00910 -0.22723 -0.01018 -0.00644
##          (0.04899) (0.17851) (0.02702) (0.02894)
## ldensity 0.13941 0.22562 * 0.42903 *** 0.43434 ***
##          (1.02124) (0.10247) (0.05485) (0.07115)
## lwcon -0.02873 0.31400 -0.00748 -0.00430
##          (0.05351) (0.25910) (0.03958) (0.04142)
## lwtuc 0.03913 -0.19894 0.04545 * 0.04446 *
##          (0.03086) (0.19712) (0.01979) (0.02154)
## lwtrd -0.01775 0.05356 -0.00814 -0.00856
##          (0.04531) (0.29600) (0.04138) (0.04198)
## lwfir -0.00934 0.04170 -0.00364 -0.00403
##          (0.03655) (0.30562) (0.02892) (0.02946)
## lwser 0.01859 -0.13543 0.00561 0.01056
##          (0.03882) (0.17365) (0.02013) (0.02158)
## lwmfg -0.24317 -0.04200 -0.20414 * -0.20180 *
##          (0.41955) (0.15627) (0.08044) (0.08394)
## lwfed -0.45134 0.14803 -0.16351 -0.21346
##          (0.52712) (0.32565) (0.15945) (0.21510)
## lwsta -0.01875 -0.20309 -0.05405 -0.06012
##          (0.28082) (0.29815) (0.10568) (0.12031)
## lwloc 0.26326 0.04444 0.16305 0.18354
##          (0.31239) (0.49436) (0.11964) (0.13968)
## lpctymle 0.35112 -0.09472 -0.10811 -0.14587
##          (1.01103) (0.19180) (0.13969) (0.22681)
## smsayes -0.08050 -0.22515 -0.25955
##          (0.14423) (0.11563) (0.14997)
## lpctmin 0.16890 ** 0.18904 *** 0.19488 ***
##          (0.05270) (0.04150) (0.04594)
## (Intercept) -1.97714 -0.95380 -0.45385
##          (4.00081) (1.28397) (1.70298)
## -----
## R^2 0.44364 0.87385 0.59847 0.59230
## Adj. R^2 0.32442 0.83729 0.58115 0.57472
## Num. obs. 630 90 630 630
## s_idios 0.14924 0.14924
## s_id 0.21456 0.21456
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

The Hausman-Taylor model (Hausman and Taylor (1981)) may be estimated with the `plm` function by setting argument `random.method = "ht"` and `inst.method = "baltagi"`. The following example is from B. H. Baltagi (2005), p. 130; B. H. Baltagi (2013), pp. 145-7, tables 7.4-7.6; B. H. Baltagi (2021), pp. 174-6, tables 7.5-7.7.

```
data("Wages", package = "plm")
ht <- plm(lwage ~ wks + south + smsa + married + exp + I(exp^2) +
          bluecol + ind + union + sex + black + ed |
          bluecol + south + smsa + ind + sex + black |
          wks + married + exp + I(exp^2) + union,
          data = Wages, index = 595,
```



```

inst.method = "baltagi", model = "random",
random.method = "ht")

am <- update(ht, inst.method = "am")
bms <- update(ht, inst.method = "bms")
screenreg(list("Hausman-Taylor" = ht, "Amemiya-MaCurdy" = am,
              "Breusch-Mizon-Schmidt" = bms),
          digits = 5, single.row = FALSE)

##
## =====
##           Hausman-Taylor  Amemiya-MaCurdy  Breusch-Mizon-Schmidt
## -----
## (Intercept)      2.91273 ***      2.92734 ***      1.97944 ***
##                  (0.28365)      (0.27513)      (0.26724)
## wks              0.00084          0.00084          0.00080
##                  (0.00060)      (0.00060)      (0.00060)
## southyes         0.00744          0.00728          0.01467
##                  (0.03196)      (0.03194)      (0.03188)
## smsyes          -0.04183 *        -0.04195 *        -0.05204 **
##                  (0.01896)      (0.01895)      (0.01891)
## marriedyes      -0.02985          -0.03009          -0.03926 *
##                  (0.01898)      (0.01897)      (0.01892)
## exp             0.11313 ***      0.11297 ***      0.10867 ***
##                  (0.00247)      (0.00247)      (0.00246)
## exp^2           -0.00042 ***      -0.00042 ***      -0.00049 ***
##                  (0.00005)      (0.00005)      (0.00005)
## bluecolyes      -0.02070          -0.02085          -0.01539
##                  (0.01378)      (0.01377)      (0.01374)
## ind             0.01360          0.01363          0.01902
##                  (0.01524)      (0.01523)      (0.01520)
## unionyes        0.03277 *         0.03248 *         0.03786 *
##                  (0.01491)      (0.01489)      (0.01486)
## sexfemale       -0.13092          -0.13201          -0.18027
##                  (0.12666)      (0.12660)      (0.12639)
## blackyes        -0.28575          -0.28590          -0.15636
##                  (0.15570)      (0.15549)      (0.15506)
## ed              0.13794 ***      0.13720 ***      0.22066 ***
##                  (0.02125)      (0.02057)      (0.01985)
## -----
## s_idios         0.15180          0.15180          0.15180
## s_id            0.94180          0.94180          0.94180
## R^2             0.60945          0.60948          0.60686
## Adj. R^2        0.60833          0.60835          0.60572
## Num. obs.       4165            4165            4165
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05

```

Nested error component model

This section shows how the nested error component model as per B. H. Baltagi, Song, and Jung (2001) can be estimated. The model is given by :

$$y_{nt} = \alpha + \beta^T x_{jnt} + u_{jnt} = \alpha + \beta^T x_{jnt} + \mu_j + v_{jn} + \epsilon_{jnt}$$

where n and t are the individual and time indexes and j is the group index in which the individuals are nested. The error u_{jnt} consists of three components :

- μ_j is the group effect,
- v_{jn} the nested effect of the individual nested in group j
- ϵ_{jnt} is the idiosyncratic error.

In the estimated examples below (replication of B. H. Baltagi, Song, and Jung (2001), p. 378, table 6; B. H. Baltagi (2021), p. 248, table 9.1), states are nested within regions. The group index is given in the 3rd position of the `index` argument to `pdata.frame` or to `plm` directly and `plm`'s argument `effect` is set to "nested":

```
data("Produc", package = "plm")
swar <- plm(form <- log(gsp) ~ log(pc) + log(emp) + log(hwy) + log(water) + log(util) +
  unemp,
  Produc, index = c("state", "year", "region"), model = "random", effect =
  "nested", random.method = "swar")
walhus <- update(swar, random.method = "walhus")
amem <- update(swar, random.method = "amemiya")
screenreg(list("Swamy-Arora" = swar, "Wallace-Hussain" = walhus, "Amemiya" = amem), digits
  = 5)
```

```
##
## =====
##              Swamy-Arora      Wallace-Hussain  Amemiya
## -----
## (Intercept)    2.08921 ***      2.08165 ***      2.13133 ***
##                (0.14570)        (0.15035)        (0.16014)
## log(pc)         0.27412 ***      0.27256 ***      0.26448 ***
##                (0.02054)        (0.02093)        (0.02176)
## log(emp)        0.73984 ***      0.74164 ***      0.75811 ***
##                (0.02575)        (0.02607)        (0.02661)
## log(hwy)        0.07274 ***      0.07493 ***      0.07211 **
##                (0.02203)        (0.02235)        (0.02363)
## log(water)      0.07645 ***      0.07639 ***      0.07616 ***
##                (0.01386)        (0.01387)        (0.01402)
## log(util)      -0.09437 ***      -0.09523 ***      -0.10151 ***
##                (0.01677)        (0.01677)        (0.01705)
## unemp          -0.00616 ***      -0.00615 ***      -0.00584 ***
##                (0.00090)        (0.00091)        (0.00091)
## -----
## s_idios         0.03676          0.03762          0.03676
## s_id            0.06541          0.06713          0.08306
## s_gp            0.03815          0.05239          0.04659
## R^2             0.97387          0.97231          0.96799
## Adj. R^2        0.97368          0.97210          0.96776
## Num. obs.       816             816             816
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

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